中国矿业大学计算机学院

2023 级本科生课程作业 1

课程名称 Linux 操作系统

作业内容 Linux 各类基本操作命令

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LINUX 平台下的基本命令

1. 文件管理

1.1 ls 命令 – 显示目录中文件及其属性信息

显示目录中的文件及其属性信息，如下是~下的文件

```
[root@localhost ~]# ls
anaconda-ks.cfg  initial-setup-ks.cfg
```

使用-l 可以输出文件名和属性信息

```
[root@localhost ~]# ls -l
total 8
-rw-------. 1 root root 1269 Aug 26 06:32 anaconda-ks.cfg
-rw-r--r--. 1 root root 1320 Aug 25 22:33 initial-setup-ks.cfg
```

1.2 cp 命令 – 复制文件或目录

使用 cp 命令把 redis 文件复制到 dir1 目录下

```
[jaydon@localhost Desktop]$ ls
dir1  dir2  redis-6.2.6.tar.gz
[jaydon@localhost Desktop]$ cp redis-6.2.6.tar.gz ./dir1
[jaydon@localhost Desktop]$ ls ./dir1
redis-6.2.6.tar.gz
```

1.3 mkdir 命令 – 创建目录文件

在 Desktop 目录下创建 dir3 目录

```
[jaydon@localhost Desktop]$ mkdir dir3
[jaydon@localhost Desktop]$ ls
dir1  dir2  dir3  redis-6.2.6.tar.gz
```

1.4 tar 命令 – 压缩和解压缩文件

将 redis.tar.gz 文件解压到 dir2 目录中

```
[jaydon@localhost Desktop]$ tar xvf redis-6.2.6.tar.gz -C ./dir2
redis-6.2.6/
redis-6.2.6/.github/
redis-6.2.6/.github/ISSUE_TEMPLATE/
redis-6.2.6/.github/ISSUE_TEMPLATE/bug_report.md
redis-6.2.6/.github/ISSUE_TEMPLATE/crash_report.md
redis-6.2.6/.github/ISSUE_TEMPLATE/feature_request.md
redis-6.2.6/.github/ISSUE_TEMPLATE/other_stuff.md
redis-6.2.6/.github/ISSUE_TEMPLATE/question.md
redis-6.2.6/.github/workflows/
redis-6.2.6/.github/workflows/ci.yml
redis-6.2.6/.github/workflows/daily.yml
redis-6.2.6/.gitignore
redis-6.2.6/00-RELEASENOTES
redis-6.2.6/BUGS
redis-6.2.6/CONDUCT
redis-6.2.6/CONTRIBUTING
redis-6.2.6/COPYING
redis-6.2.6/INSTALL
```

```
[jaydon@localhost Desktop]$ ls ./dir2
redis-6.2.6
```

1.5 cd 命令 – 切换目录

从 Desktop 切换到./Desktop/dir2 目录中

```
[jaydon@localhost Desktop]$ cd ./dir2
[jaydon@localhost dir2]$
```

1.6 chmod 命令 – 改变文件或目录权限

将 hello.txt 设置为只读，用 ls -l 查看其权限情况

```
[jaydon@localhost dir3]$ ls -l
total 4
-rw-rw-r--. 1 jaydon jaydon 39 Nov  9 17:43 hello.txt
[jaydon@localhost dir3]$ chmod 444 hello.txt
[jaydon@localhost dir3]$ ls -l
total 4
-r--r--r--. 1 jaydon jaydon 39 Nov  9 17:43 hello.txt
```

2. 文档编辑

2.1 cat 命令 – 在终端设备上显示文件内容

```
[jaydon@localhost dir3]$ cat hello.txt
Hello. I am Wang Zhihao from DataSci-1
```

2.2 rm 命令 – 删除文件或目录

强制删除 dir2 目录中的 redis-6.2.6 文件，用 ls 查看，文件被成功删

除

```
[jaydon@localhost dir3]$ rm -rf ~/Desktop/dir2/redis-6.2.6
[jaydon@localhost dir3]$ ls ~/Desktop/dir2/
[jaydon@localhost dir3]$
```

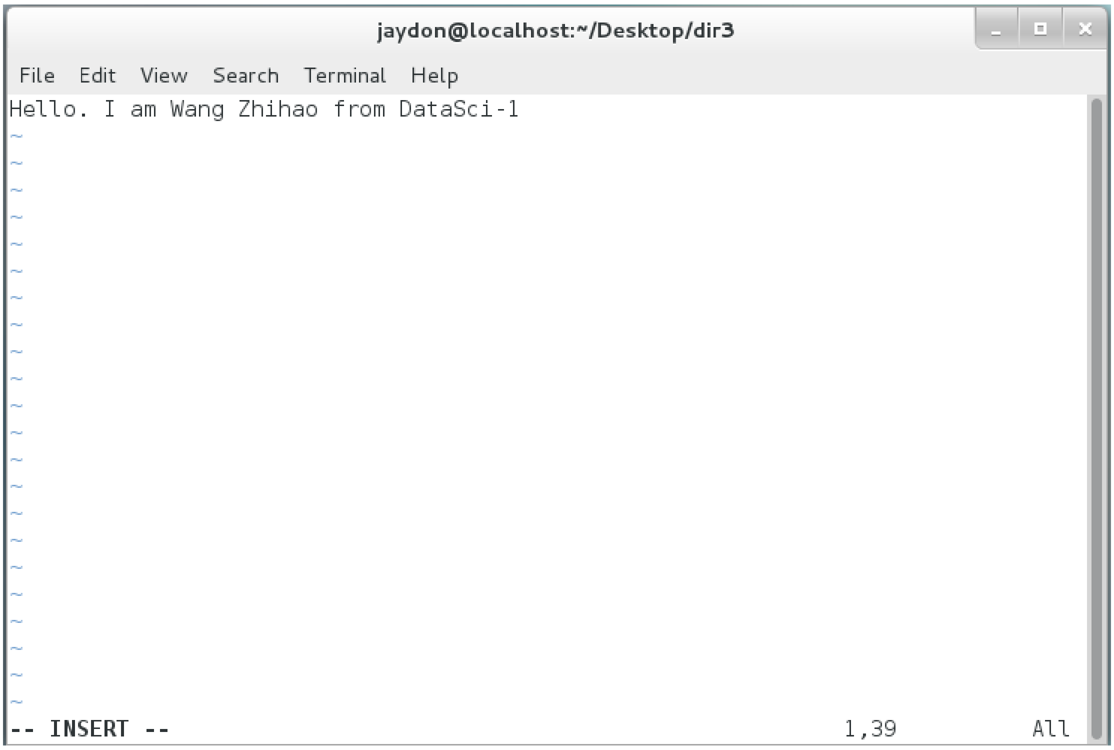
2.3 grep 命令 – 文本搜索

搜索指定文件中包含某个关键词的内容行，如下搜索 passwd 中包含 root 的内容行

```
[jaydon@localhost dir3]$ grep root /etc/passwd
root:x:0:0:root:/root:/bin/bash
operator:x:11:0:operator:/root:/sbin/nologin
```

2.4 vi 命令 – 文本编辑器

```
[jaydon@localhost dir3]$ vi hello.txt
```



2.5 echo 命令 – 输出字符串或提取后的变量值

输出指定字符串到终端设备界面

```
[jaydon@localhost dir3]$ echo wangzhihao
wangzhihao
```

搭配输出重定向符一起使用，将字符串内容直接写入文件中，下图中，

使用 echo 命令将该内容写入 old.txt 文件中，old.txt 文件本不存在，系统会自动创建并写入。

```
[jaydon@localhost dir3]$ echo "I am 21 years old." > old.txt
[jaydon@localhost dir3]$ ls
hello.txt  old.txt
[jaydon@localhost dir3]$ cat old.txt
I am 21 years old.
```

3. 系统管理

3.1 find 命令 – 根据路径和条件搜索指定文件

在某目录下搜索 txt 后缀的文件，结果如下

```
[jaydon@localhost dir3]$ find /home/jaydon/Desktop/ *.txt
/home/jaydon/Desktop/
/home/jaydon/Desktop/redis-6.2.6.tar.gz
/home/jaydon/Desktop/dir1
/home/jaydon/Desktop/dir1/redis-6.2.6.tar.gz
/home/jaydon/Desktop/dir2
/home/jaydon/Desktop/dir3
/home/jaydon/Desktop/dir3/hello.txt
/home/jaydon/Desktop/dir3/old.txt
hello.txt
old.txt
```

在某目录下搜索权限为只读的文件，结果如下

```
[jaydon@localhost dir3]$ find /home/jaydon/Desktop/ -perm 444
/home/jaydon/Desktop/dir3/hello.txt
```

3.2 ps 命令 – 显示进程状态

将系统运行状态中指定用户的进程信息过滤出来

```
[jaydon@localhost dir3]$ ps -u jaydon
   PID TTY          TIME CMD
  2549 ?        00:00:00 gnome-keyring-d
  2551 ?        00:00:00 gnome-session
  2559 ?        00:00:00 dbus-launch
  2560 ?        00:00:00 dbus-daemon
  2625 ?        00:00:00 gvfsd
  2631 ?        00:00:00 gvfsd-fuse
  2703 ?        00:00:00 ssh-agent
  2724 ?        00:00:00 at-spi-bus-laun
```

3.3 kill 命令 – 杀死进程

先创建出一个文本编辑器进程，然后再强制将该进程杀死

```
[jaydon@localhost dir3]$ gedit &
[1] 40755
[jaydon@localhost dir3]$ ps aux | grep gedit
jaydon    40755  1.2  0.5 728484 22720 pts/1    Sl   23:36   0:00 gedit
jaydon    40768  0.0  0.0 112808   980 pts/1    S+   23:37   0:00 grep --color=a
uto gedit
[jaydon@localhost dir3]$ kill -9 40755
[1]+  Killed                  gedit
```

3.4 uname 命令 – 显示系统内核信息

输入 uname -a 命令，可以显示系统所有相关信息（含内核名称、主机名、版本号及硬件架构）

```
[jaydon@localhost dir3]$ uname -a
Linux localhost.localdomain 3.10.0-123.el7.x86_64 #1 SMP Mon Jun 30 12:09:22 UTC
 2014 x86_64 x86_64 x86_64 GNU/Linux
```

3.5 rpm 命令 – RPM 软件包管理器

使用 rpm -ivh 命令将 mysql.rpm 安装，并且通过 rpm -qa | grep mysql80 的命令，可以看到该安装包成功安装

```
[root@localhost ~]# rpm -ivh /tmp/mysql80-community-release-el7-6.noarch.rpm
warning: /tmp/mysql80-community-release-el7-6.noarch.rpm: Header V4 RSA/SHA256 Signature, key ID 3a79bd29: NOKEY
Preparing...                          ################################# [100%]
Updating / installing...
   1:mysql80-community-release-el7-6  ################################# [100%]
You have new mail in /var/spool/mail/root
[root@localhost ~]# rpm -qa | grep mysql80
mysql80-community-release-el7-6.noarch
```

通过-ql 的选项可以观察到软件包的安装路径

```
[root@localhost ~]# rpm -ql mysql80-community-release-el7-6.noarch
/etc/RPM-GPG-KEY-mysql
/etc/RPM-GPG-KEY-mysql-2022
/etc/yum.repos.d/mysql-community-debuginfo.repo
/etc/yum.repos.d/mysql-community-source.repo
/etc/yum.repos.d/mysql-community.repo
```

4. 磁盘管理

4.1 df 命令 – 显示磁盘空间使用量情况

输入 df 即可查看磁盘空间使用情况

```
[root@localhost ~]# df
Filesystem              1K-blocks     Used Available Use% Mounted on
/dev/mapper/centos-root  39167300  5177280  33990020  14% /
devtmpfs                  1925256        0   1925256   0% /dev
tmpfs                     1934520      148   1934372   1% /dev/shm
tmpfs                     1934520     9076   1925444   1% /run
tmpfs                     1934520        0   1934520   0% /sys/fs/cgroup
/dev/mapper/centos-home  19122176    75780  19046396   1% /home
/dev/sda1                  508588   121200    387388  24% /boot
/dev/sr0                 10540998 10540998         0 100% /run/media/jaydon/CentOS-8-5-2111-x86_64-dvd
```

4.2 lsblk 命令 – 查看系统的磁盘使用情况

输入 lsblk 即可查看系统的磁盘使用情况

```
[root@localhost ~]# lsblk
NAME            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda               8:0    0   60G  0 disk
├─sda1            8:1    0  500M  0 part /boot
└─sda2            8:2    0 59.5G  0 part
  ├─centos-swap 253:0    0  3.9G  0 lvm  [SWAP]
  ├─centos-root 253:1    0 37.4G  0 lvm  /
  └─centos-home 253:2    0 18.3G  0 lvm  /home
sr0              11:0    1 10.1G  0 rom  /run/media/jaydon/CentOS-8-5-2111-x86_64-dvd
```

4.3 fdisk 命令 – 管理磁盘分区

使用 fdisk -l 命令显示当前系统分区情况

```
[root@localhost ~]# fdisk -l

Disk /dev/sda: 64.4 GB, 64424509440 bytes, 125829120 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x0008f1f4

   Device Boot      Start         End      Blocks   Id  System
/dev/sda1   *        2048     1026047      512000   83  Linux
/dev/sda2         1026048   125829119    62401536   8e  Linux LVM

Disk /dev/mapper/centos-swap: 4177 MB, 4177526784 bytes, 8159232 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/mapper/centos-root: 40.1 GB, 40126906368 bytes, 78372864 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/mapper/centos-home: 19.6 GB, 19591593984 bytes, 38264832 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

5. 网络通讯

5.1 ifconfig 命令 – 显示或设置网络设备

使用 ifconfig 命令显示当前网络设备情况

```
[root@localhost ~]# ifconfig
eno16777736: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.244.131  netmask 255.255.255.0  broadcast 192.168.244.255
        inet6 fe80::20c:29ff:fe2e:8d55  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:2e:8d:55  txqueuelen 1000  (Ethernet)
        RX packets 883949  bytes 809367389 (771.8 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 666177  bytes 69383719 (66.1 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 0  (Local Loopback)
        RX packets 4101  bytes 216589 (211.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4101  bytes 216589 (211.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

还可以对指定的网卡设备执行 IP 地址修改操作，操作与结果如下：

```
[root@localhost ~]# ifconfig eno16777736 192.168.10.20 netmask 255.255.255.0
[root@localhost ~]# ifconfig
eno16777736: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.10.20  netmask 255.255.255.0  broadcast 192.168.10.255
        inet6 fe80::20c:29ff:fe2e:8d55  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:2e:8d:55  txqueuelen 1000  (Ethernet)
        RX packets 883971  bytes 809369726 (771.8 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 666189  bytes 69386591 (66.1 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

5.2 ping 命令 – 检测与另一个主机之间的网络连接

尝试 ping baidu.com，发现可以 ping 通，说明连通性正常

```
[root@localhost ~]# ping www.baidu.com
PING www.a.shifen.com (36.152.44.93) 56(84) bytes of data.
64 bytes from 36.152.44.93: icmp_seq=1 ttl=128 time=46.3 ms
64 bytes from 36.152.44.93: icmp_seq=2 ttl=128 time=76.6 ms
64 bytes from 36.152.44.93: icmp_seq=3 ttl=128 time=55.2 ms
64 bytes from 36.152.44.93: icmp_seq=4 ttl=128 time=51.4 ms
64 bytes from 36.152.44.93: icmp_seq=5 ttl=128 time=68.1 ms
64 bytes from 36.152.44.93: icmp_seq=6 ttl=128 time=61.7 ms
64 bytes from 36.152.44.93: icmp_seq=7 ttl=128 time=66.7 ms
64 bytes from 36.152.44.93: icmp_seq=8 ttl=128 time=58.8 ms
^C
--- www.a.shifen.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7011ms
rtt min/avg/max/mdev = 46.321/60.634/76.622/9.165 ms
```

5.3 ssh 命令 – 安全的远程连接服务

我们使用 windows 的 cmd 来进行 ssh 连接其中一台虚拟机，操作如

下，可以看到我们的本地端口是 9000，目标主机是 localhost，访问的目标端口是 8081，访问的用户是 wang，目标 ip 地址是 192.168.250.240，在连接成功后可以正常执行 linux 操作

```
C:\Users\Lenovo>ssh -L 9000:localhost:8081 wang@192.168.250.240
The authenticity of host '192.168.250.240 (192.168.250.240)' can't be established.
ED25519 key fingerprint is SHA256:zU+30emRBLSxKMTGA/UuNl7d5RkCmodmcuo4eaFT72U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.250.240' (ED25519) to the list of known hosts.
wang@192.168.250.240's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sun Nov  9 08:12:43 2025
[wang@localhost ~]$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
[wang@localhost ~]$
```

5.4 netstat 命令 – 显示网络状态

如下图所示，使用 netstat 命令可以显示所有的网络状态

```
[root@localhost ~]# netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ]         DGRAM                    12558    /run/systemd/shutdownd
unix  6      [ ]         DGRAM                    8008     /run/systemd/journal/socket
unix  28     [ ]         DGRAM                    8010     /dev/log
unix  2      [ ]         DGRAM                    13401    @/org/freedesktop/systemd1/notify
unix  3      [ ]         STREAM     CONNECTED     17154
unix  3      [ ]         STREAM     CONNECTED     27497    /var/run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     23556
unix  3      [ ]         STREAM     CONNECTED     26073
unix  3      [ ]         STREAM     CONNECTED     666477   @/tmp/.X11-unix/X0
```

6. 结合管道操作符组合命令

6.1 查找 jaydon 用户的进程中带有 gvf 内容的进程：

```
[jaydon@localhost dir3]$ ps -u jaydon | grep gvf*
  2625 ?        00:00:00 gvfsd
  2631 ?        00:00:00 gvfsd-fuse
  2779 ?        00:00:00 gvfs-udisks2-vo
  2794 ?        00:00:00 gvfs-afc-volume
  2801 ?        00:00:00 gvfs-mtp-volume
  2807 ?        00:00:00 gvfs-goa-volume
  2819 ?        00:00:00 gvfs-gphoto2-vo
  3043 ?        00:00:00 gvfsd-trash
  3108 ?        00:00:00 gvfsd-metadata
 38145 ?        00:00:00 gvfsd-recent
```

6.2 查看占用内存最高的进程：

ps aux：列出所有进程

sort -nrk 3：按第 3 列（CPU 占用率）从大到小排序

head：显示前 10 个

```
[root@localhost ~]# ps aux | sort -nrk 3 | head
jaydon     2822  1.2 17.0 2216048 657920 ?      Sl   Nov09  16:11 /usr/bin/gnome-shell
root        970  0.2  0.0   4360   596 ?        Ss   Nov09   3:45 /sbin/rngd -f
root        139  0.2  0.0      0     0 ?        R    Nov09   3:03 [rcuos/1]
root        138  0.1  0.0      0     0 ?        S    Nov09   2:21 [rcuos/0]
root        137  0.1  0.0      0     0 ?        S    Nov09   2:13 [rcu_sched]
root       1302  0.1  0.0 145708  3428 ?        Ssl  Nov09   2:16 /usr/local/bin/redis-server 0.0.0.0:6379
root       1070  0.1  1.4 226244 54812 tty1     Rsl+ Nov09   1:22 /usr/bin/Xorg :0 -background none -verbose
r-gdm-hRroUF/database -seat seat0 -nolisten tcp vt1
USER        PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
rtkit       981  0.0  0.0 164704  1284 ?        SNsl Nov09   0:02 /usr/libexec/rtkit-daemon
rpcuser    1354  0.0  0.0  46644  1836 ?        Ss   Nov09   0:00 /sbin/rpc.statd
```

6.3 统计文件行数、单词数、字符数：

等价于 wc hello.txt

```
[jaydon@localhost dir3]$ cat hello.txt | wc
      1       7      39
```

6.4 查看文件中出现次数最多的单词：

tr -s ' ' '\n'：把空格换成换行，一行一个单词

uniq -c：统计重复次数

sort -nr：按次数从大到小排序

head：显示前 10 个

```
[jaydon@localhost dir3]$ cat hello.txt | tr -s ' ' '\n' | sort | uniq -c | sort
-nr | head
      1 Zhihao
      1 Wang
      1 I
      1 Hello.
      1 from
      1 DataSci-1
      1 am
```

6.5 综合查看 CPU、内存、磁盘、网络实时状态：

```
[jaydon@localhost dir3]$ echo -e "CPU:\n$(top -bn1 | grep 'Cpu(s)')" \
> "\nMEM:\n$(free -h | grep Mem:)" \
> "\nDISK:\n$(df -h | grep '/$')" \
> "\nNET:\n$(ss -s)"
CPU:
%Cpu(s):  0.7 us,  0.6 sy,  0.0 ni, 98.7 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MEM:
Mem:          3.7G       2.9G       791M        10M       2.3M       1.6G
DISK:
/dev/mapper/centos-root   38G  5.0G   33G  14% /
NET:
Total: 1127 (kernel 0)
TCP:   12 (estab 0, closed 1, orphaned 0, synrecv 0, timewait 0/0), ports 0

Transport Total     IP        IPv6
*         0         -         -
RAW       1         0         1
UDP       19        12        7
TCP       11        6         5
INET      31        18        13
FRAG      0         0         0
```

6.6 找出系统中体积最大的前 10 个文件:

```
[root@localhost ~]# find / -type f -exec du -h {} + 2>/dev/null | sort -hr | head
681M    /run/media/jaydon/CentOS-8-5-2111-x86_64-dvd/images/install.img
318M    /run/media/jaydon/CentOS-8-5-2111-x86_64-dvd/AppStream/Packages/java-11-openjdk-jmods-11.0.13.0.8-1.el8_4.x86_64.rpm
264M    /run/media/jaydon/CentOS-8-5-2111-x86_64-dvd/AppStream/Packages/qt5-qt3d-examples-5.15.2-2.el8.x86_64.rpm
239M    /run/media/jaydon/CentOS-8-5-2111-x86_64-dvd/AppStream/Packages/java-17-openjdk-jmods-17.0.0.0.35-4.el8.x86_64.rpm
226M    /run/media/jaydon/CentOS-8-5-2111-x86_64-dvd/AppStream/Packages/mysql-test-8.0.26-1.module_el8.4.0+915+de215114.x86_64.rpm
180M    /run/media/jaydon/CentOS-8-5-2111-x86_64-dvd/AppStream/Packages/virtio-win-1.9.19-1.el8.noarch.rpm
162M    /run/media/jaydon/CentOS-8-5-2111-x86_64-dvd/BaseOS/Packages/linux-firmware-20210702-103.gitd79c2677.el8.noarch.rpm
156M    /var/cache/yum/x86_64/7/updates/gen/primary_db.sqlite
149M    /var/cache/yum/x86_64/7/updates/gen/filelists_db.sqlite
117M    /var/cache/yum/x86_64/7/updates/packages/firefox-115.12.0-1.el7.centos.x86_64.rpm
```

find / -type f: 全系统查找文件

du -h: 显示可读大小

2>/dev/null: 屏蔽无权限报错

sort -hr: 按大小排序（从大到小）

head: 看前 10

个人体会:

完成本次 Linux 基本命令作业让我对操作系统有了更深入的理解。从最初对命令行界面的陌生到能够熟练运用各种命令解决实际问题，这个过程充满了挑战与收获。通过实践，我认识到 Linux 命令的强大之处在于它们的组合使用，管道操作符让简单的命令能够完成复杂的

任务。特别是在学习权限管理和进程控制时，我体会到了 Linux 系统安全设计的精妙之处。每个命令看似简单，但背后的原理和适用场景却值得深入思考。这次作业不仅让我掌握了基本的文件操作、系统管理和网络配置技能，更重要的是培养了我通过命令行与计算机系统高效交互的思维方式。这些基础技能为我后续学习更高级的系统管理和 Shell 编程打下了坚实的基础。